

# Selection rules for the $N = 2$ flow $k \rightarrow k - 2$ with superfields as building blocks, and the massive vacua of the massless phase

Working note

## 1 Rule: nodes are made of full superfields

A superprimary and its superdescendants must enter an algebra together. The building blocks are therefore not the sectors  $(l, m, s)$  of the bosonic subalgebra, but the NS superfields

$$[l, m] \equiv [(l, m, 0) \oplus (l, m, 2)] \otimes \overline{[(l, -m, 0) \oplus (l, -m, 2)]}, \quad l+m \text{ even}, \quad [l, m] \sim [k-l, m+K], \quad (1)$$

with  $K = k + 2$ . Each superfield contains all its components: the bottom component, the  $G_{-1/2}$ ,  $\bar{G}_{-1/2}$  and  $G_{-1/2}\bar{G}_{-1/2}$  descendants, and the D-term. The fermionic mixed components are included, so the chiral algebra is the full  $N = 2$  algebra with  $G^\pm$ .

In the bosonic diagonal model of N2Gab (2.5) the left-only descendants  $(l, m, 2) \otimes \overline{(l, -m, 0)}$  are absent. That is why the sector-by-sector analysis of the previous versions was not a supersymmetric one.

For odd  $k$  every superfield has a unique representative with  $l$  even, and then  $m$  is even mod  $2K$ . The superfield fusion is  $su(2)_k$  on  $l$  and addition on  $m$ , so

$$\{\text{NS superfields}\} \simeq so(3)_k \times \mathbb{Z}_K, \quad [l, m] \leftrightarrow (l \in so(3)_k, m/2 \in \mathbb{Z}_K), \quad d_{[l, m]} = d_l. \quad (2)$$

There are  $(k+1)(k+2)/2$  superfields. The Verlinde lines that commute with the  $N = 2$  algebra are labelled by the same set. The line  $L_{[L, M]}$  acts on  $[l, m]$  uniformly on all components, by  $\frac{S_{Ll}}{S_{0l}} e^{i\pi M m/K}$ . Here:

- $L_{[L, 0]}$  ( $L$  even) are the  $so(3)_k$  lines;
- $L_{[0, 2n]}$  is the phase symmetry  $X \rightarrow \omega^n X$  (N2Gab §5.4).

A node  $T$ , its commuting lines  $F_T$  and the global index  $\mu_T = (\sum_{F_T} d^2)^2$  are defined as in selection.pdf and paul.pdf, now with superfields.

*What drops out.* In the sector-by-sector analysis we found extra lines,  $L_{(0,0,2)}$  and  $L_{(0,K,\pm 1)}$ . They act differently on the components of one superfield:  $L_{(0,0,2)}$  is trivial on all NS superfields, and  $L_{(0,K,\pm 1)}$  gives opposite signs on bottom and top components. So they are not lines of the supersymmetric theory.  $L_{(0,K,\pm 1)}$  reappears below as a global  $R$ -symmetry.

All numbers come from `n2.superfields.py`, checked for  $k = 3, \dots, 13$ .

## 2 UV: the smallest node containing the perturbation

The perturbation is the F-term of the chiral superfield  $[k, k] \sim [0, -2] (X^k)$ , plus its conjugate. The smallest node is

$$T_{UV} = \langle [k, k] \rangle = \{[0, 2n]\}_{n \in \mathbb{Z}_K} \simeq \mathbb{Z}_K, \quad F_{UV} = \{L_{[L, 0]}\}_{L \text{ even}} = \mathfrak{so}(3)_k, \quad \mu_{UV} = (\dim \mathfrak{so}(3)_k)^2, \quad (3)$$

with  $\dim \mathfrak{so}(3)_k = \sum_{L \text{ even}} d_L^2 = K/(4 \sin^2 \frac{\pi}{K})$ . In standard labels ( $|m| \leq l$ ),  $T_{UV}$  contains the superprimaries

$$[0, 0], \quad [k, m], \quad m = -k, -k+2, \dots, k, \quad h = \frac{k(k+2) - m^2}{4K}, \quad q = \frac{m}{K}. \quad (4)$$

| $k$      | relevant, $N = 2$ -preserving                        | relevant components that break $N = 2$             |
|----------|------------------------------------------------------|----------------------------------------------------|
| 3        | F-terms of $[3, \pm 3]$ , $\Delta = 8/5$             | bottoms of $[3, \pm 3]$ (3/5), $[3, \pm 1]$ (7/5)  |
| 5        | F-terms of $[5, \pm 5]$ , $\Delta = 12/7$            | bottoms of $[5, \pm 5]$ (5/7), $[5, \pm 3]$ (13/7) |
| $\geq 7$ | F-terms of $[k, \pm k]$ , $\Delta = 2 - \frac{2}{K}$ | bottoms of $[k, \pm k]$ ( $k/K$ )                  |

D-terms have  $\Delta = 2h + 2 > 2$  and are never relevant. So  $\phi_{UV}$  is the only  $N = 2$ -preserving relevant perturbation in  $T_{UV}$ , and it is exclusive to it.

Every nontrivial line of  $\mathfrak{so}(3)_k$  generates all of  $\mathfrak{so}(3)_k$  (checked). Hence the only nodes containing  $T_{UV}$  are  $T_{UV}$  ( $\mu = \dim^2 \mathfrak{so}(3)_k$ ) and the complete theory  $C_k$  ( $\mu = 1$ ).  $C_k$  contains the F-terms of all  $X^j$ , and those with  $j < k$  are exclusive to it.

## 3 IR: the node of the IR operator

The  $(k-2)$  model is approached by the D-term of the non-BPS superfield  $[4, \pm 2]$  (N2Gab §5.3,  $\Delta = 2 + 10/k$ ). For  $k' = k-2 \geq 5$ :

$$\langle [4, 2] \rangle_{k-2} = C_{k-2}, \quad F = \{1\}, \quad \mu_{IR} = 1. \quad (5)$$

The reason:  $l = 4$  generates all of  $\mathfrak{so}(3)_{k-2}$ , and  $m = 2$  generates  $\mathbb{Z}_{K-2}$ . For  $k \leq 5$  the field  $[4, 2]$  does not exist, but any IR superfield with  $l \neq 0$  and  $m \neq 0$  again generates  $C_{k-2}$ .

| $k$ | $\dim \mathfrak{so}(3)_k$ | $\mu(T_{UV})$ | $\mu(\langle [4, 2] \rangle_{k-2})$ |
|-----|---------------------------|---------------|-------------------------------------|
| 3   | 3.618                     | 13.09         | —                                   |
| 5   | 9.296                     | 86.41         | —                                   |
| 7   | 19.23                     | 369.96        | 1                                   |
| 9   | 34.65                     | 1200.38       | 1                                   |
| 11  | 56.75                     | 3220.20       | 1                                   |
| 13  | 86.75                     | 7525.70       | 1                                   |

## 4 Selection rules

**(a)  $X^k$  alone.** The flow stays in  $T_{UV}$ , and  $\mathfrak{so}(3)_k$  is preserved at all scales (transportability, selection.pdf §3). The IR nodes available are  $C_{k-2}$  only, with  $\mu = 1 \neq \dim^2 \mathfrak{so}(3)_k$ . So there is no massless IR, as in item (d) of selection.pdf §3.

This holds even allowing extra gapped vacua. A line cannot connect the  $c_{k-2} > 0$  vacuum to a gapped one, so it acts on the CFT vacuum by a number  $\chi(L) = \sum n_j d_j^{IR} > 0$ . Positivity forces

$\chi = d_L$ . But  $d_L \in \mathbb{Q}(\zeta_K)$  and  $d_j^{IR} \in \mathbb{Q}(\zeta_{8(K-2)})$ , whose real intersection is  $\mathbb{Q}$  for odd  $K$ , while  $d_2 = 1 + 2 \cos \frac{2\pi}{K} \notin \mathbb{Q}$  for  $K \geq 5$ .

The flow is therefore fully massive. It is the Chebyshev deformation, the unique superpotential keeping  $\mathfrak{so}(3)_k$  (N2Gab §4).

**(b) The massless flow.** The couplings of  $X^k$  and  $X^{k-2}$  have dimensions  $2/K$  and  $4/K = 2 \cdot \frac{2}{K}$ , so they are resonant. At  $O(\lambda^2)$  the linear deformation  $X^{k+2} + \lambda X^k$  therefore also turns on the F-term of  $[k-2, k-2] \sim [2, -4]$ . This superfield is exclusive to  $C_k$ , and it is where the second-order obstruction of N2Gab §3.4 sits. The node becomes

$$C_k \longrightarrow C_{k-2}, \quad \mu = 1 = 1. \quad (6)$$

The selection rule is satisfied, but it is empty: *no line commuting with  $N = 2$  survives the massless flow.*

## 5 Entropic order parameters and the massive vacua

**Language.** For a pair of algebras  $O \subset M$  on a region  $R$  with index  $\lambda$ , the certainty relation (entropic.pdf (3.25)) reads

$$S_{M(R)}(\omega|\omega \circ E) + S_{O(R')}(\omega|\omega \circ E') = \log \lambda, \quad 0 \leq \cdot \leq \log \lambda. \quad (7)$$

The first term tests the charged operators (intertwiners), the second the twists.

In  $d = 2$  these two kinds of operator are not distinguished by the topology of their regions. Take a two-interval region with cross-ratio  $x$ : its complement is again two intervals, with cross-ratio  $1 - x$ . The twists of one are the intertwiners of the other, so the certainty relation just pairs  $x$  with  $1 - x$ . For a single interval  $R$  (entropic.pdf (3.58)), the complement is two half-lines, and the twist implementing the symmetry on  $R$  sits there exactly like an intertwiner of that two-component region.

The phase is read from the large-scale behaviour at fixed geometry (entropic.pdf §3.4, §4):

- *massless*: conformal functions;
- *gapped, unbroken*: charged part  $\rightarrow 0$ , twists  $\rightarrow \log \lambda$ ;
- *gapped, broken*: charged part  $\rightarrow \log \lambda_{\text{vac}}$  (constant law), with  $1 \leq \lambda_{\text{vac}} \leq \lambda$ .

Here  $\lambda_{\text{vac}} = D_{\text{TQFT}}^2$  is the total quantum dimension of the topological (vacuum-degeneracy) operators. For a group it is the number of vacua in one orbit.

**Massless flow: lines give nothing.** The node pair is  $C_k \rightarrow C_{k-2}$  with  $\lambda = 1$ , so every order parameter built from lines commuting with  $N = 2$  vanishes identically. Nothing topological is forced on top of the  $(k-2)$  model by such lines. In particular, no TQFT inside the CFT vacuum is required.

**Massless flow: the  $R$ -parity.** The complete theory has one global symmetry that commutes with the perturbation. It is the  $R$ -parity  $G_R$ :  $X \rightarrow -X$ ,  $W \rightarrow -W$ , compensated by  $\theta \rightarrow i\theta$ . On bosonic operators it is a  $\mathbb{Z}_2$ ; on fermions it squares to  $(-1)^F$ .

- It maps each superfield to itself but acts differently on its components, so it does not commute with  $N = 2$  and is not a node line. This is the only place where  $L_{(0,K,\pm 1)}$  enters.

- Take  $O = C^{G_R}$  (the  $G_R$ -even operators)  $\subset C$ , with  $\lambda = |G_R| = 2$ . The charged operators are the odd chiral fields, e.g.  $X$ . We use the single-interval order parameter  $S_{C(R)}(\omega|\omega \circ E_\psi)$  of entropic.pdf (3.58).
- *CFT vacuum*:  $X \rightarrow 0$  and  $G_R$  is unbroken, so the order parameter  $\rightarrow 0$ .
- *Massive vacua*:  $W(v) \neq 0$  and  $G_R$  flips the sign of  $W$ , so a massive vacuum is never  $G_R$ -invariant:  $\lambda_{\text{vac}} \geq 2$ . The certainty bound gives  $\lambda_{\text{vac}} \leq |G_R| = 2$ . Hence

$$D_{\text{TQFT}}^2 = 2 : \quad \langle X \rangle = \pm i\sqrt{\lambda}, \quad S_{C(R)}(\omega|\omega \circ E_\psi) \rightarrow \log 2, \quad (8)$$

with a constant law, while the twist parameter  $\rightarrow 0$ .

This uses only the certainty relation and  $W(v) \neq 0$ . It agrees with the Witten index,  $k+1 = (k-1)+2$ . The bound covers vacua related by symmetry; vacua not related by any symmetry cannot be bounded this way, and the Witten index excludes them here.

**Check with the Chebyshev flow (a).** Here  $\lambda = \dim \mathfrak{so}(3)_k \cdot |G_R|$ .

- In a gapped phase a line maps a vacuum to a superposition of vacua with non-negative integer multiplicities, and the largest eigenvalue of that matrix is  $d_L$ . Since  $d_L \notin \mathbb{Z}$ ,  $\mathfrak{so}(3)_k$  cannot be unbroken: it needs at least  $\deg \mathbb{Q}(d_2) = \varphi(K)/2$  vacua, and  $(k+1)/2$  for the regular action, which gives  $\lambda_{\text{vac}} = \dim \mathfrak{so}(3)_k$ .
- $G_R$  doubles the vacua.

The total is  $\frac{k+1}{2} \cdot 2 = k+1$ , exactly the Chebyshev vacua. The same bound shows that two massive vacua cannot host  $\mathfrak{so}(3)_k$  for  $k \geq 5$ . This is another way to see that the massless flow must break  $\mathfrak{so}(3)_k$  explicitly.

**Relation to massless.pdf.** In the massless flow massless and massive physics coexist, in different vacua. The symmetry predicted by the conjecture is  $G_R$ . The massive operators are the  $G_R$ -odd ones (e.g.  $X$ , with  $\langle X \rangle \neq 0$  in the massive vacua). In the CFT vacuum the massless algebra  $A_m$  is all of  $C_{k-2}$ .

## Summary

1. With superfields as units, the smallest algebra containing  $(0, -2, 0)$  is  $\{[0, 2n]\} \simeq \mathbb{Z}_K$ : the superprimaries  $[0, 0]$  and  $[k, m]$  ( $m$  odd). Its only  $N = 2$ -preserving relevant perturbation is the F-term of  $X^k$ .  $F = \mathfrak{so}(3)_k$  and  $\mu = \dim^2 \mathfrak{so}(3)_k$ . It has no massless IR, so it gives the Chebyshev flow with  $k+1$  vacua.
2. The massless flow needs the resonant F-term of  $X^{k-2}$ , which forces the complete node:  $C_k \rightarrow C_{k-2}$ ,  $\mu = 1$ . No line commuting with  $N = 2$  survives.
3. The TQFT on top of the  $(k-2)$  model consists of two vacua exchanged by  $R$ -parity. The entropic bound  $\lambda_{\text{vac}} \leq |G_R| = 2$  and  $W(v) \neq 0$  fix  $D_{\text{TQFT}}^2 = 2$ . There is no topological sector inside the CFT vacuum.

**Caveats.**

- Odd  $k$  only; for even  $k$  the line  $[k, 0]$  is invertible and survives (N2Gab App. D).
- “Flow inside  $T_{UV} = \text{Chebyshev}$ ” uses N2Gab §4.
- Index theory for fermionic nets is used at the level of the NS superfield fusion ring.
- The field argument assumes IR line dimensions lie in the cyclotomic field of the  $(k - 2)$  model.